

Varsha C S

Research Scholar in Mathematics | **Specialization:** Relativity & Cosmology

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CAREER OBJECTIVE

To secure a position as Assistant Professor / Guest Lecturer in the Department of Mathematics at a reputed institution, where I can apply my strong foundation in pure and applied mathematics, my active research record in General Relativity and Cosmology, and my passion for teaching to inspire students, deliver high-quality instruction, and contribute meaningfully to the academic and research culture of the department. My doctoral thesis is in the final stage of preparation, with synopsis submission scheduled for 10 June 2026 and thesis submission for 16 August 2026.

AREAS OF SPECIALIZATION & RESEARCH INTERESTS

- General Theory of Relativity and Modified Gravity Theories
- Wormhole Geometry, Black Hole Thermodynamics and Spacetime Solutions
- Cosmology, Differential Geometry, Tensor Analysis and Mathematical Physics

TEACHING INTERESTS

Competent to teach the following subjects at undergraduate (B.Sc / B.E.) and postgraduate (M.Sc) level:

- Algebra (Linear Algebra, Abstract Algebra, Group & Ring Theory)
- Real Analysis, Complex Analysis and Numerical Analysis
- Differential Geometry, Tensor Calculus and Riemannian Geometry
- Ordinary and Partial Differential Equations
- Special and General Relativity, Mathematical Physics
- Fluid Mechanics, Calculus and Engineering Mathematics

EDUCATION

Doctor of Philosophy (Ph.D.) in Mathematics | 2023 – Present

Kuvempu University, Jnanasahyadri, Shankaraghatta, Shivamogga, Karnataka, India

- Thesis Title: "Study on Relativistic Dynamics of Einstein's Solutions in Spacetime Geometry"
- **Synopsis Submission:** May 2026 | **Thesis Submission:** 16 August 2026 (scheduled)

Master of Science (M.Sc.) in Mathematics | 2020 – 2022

Kuvempu University, Jnanasahyadri, Shankaraghatta, Shivamogga, Karnataka, India

- **CGPA:** 9.13 | **Secured Second Rank in the University**
- **Key Subjects:** Algebra, Differential Geometry, Tensors, Complex & Real Analysis, Numerical Analysis, Special and General Relativity, Differential Equations, Fluid Mechanics.

Bachelor of Science (B.Sc.) – Physics, Chemistry and Mathematics (PCM) | 2017 – 2020

Sahyadri Science College, Shivamogga, Karnataka, India

- **Aggregate:** 90.5% | **Gold Medalist** (Highest marks in B.Sc. and in Mathematics)

Pre-University Course (PUC – 12th) | 2015 – 2017

Sri Vidyabharathi Independent PU College, Shivamogga, Karnataka, India

- **Combination:** PCMB (Physics, Chemistry, Mathematics, Biology)
- **Aggregate:** 96%

Secondary School Leaving Certificate (SSLC – 10th) | 2015

S. Ramaiah Sarvodaya Girls' High School, Jayanagara, Shivamogga, Karnataka, India

- **Aggregate:** 94.24%

ACADEMIC ELIGIBILITY

- **Qualified Karnataka State Eligibility Test (KSET), 2025** — eligible for the post of Assistant Professor in Universities and Colleges in Karnataka.

RESEARCH EXPERIENCE

Visiting Research Scholar | 02 – 20 April 2026

Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India

- Undertook a three-week academic visit at IUCAA, one of India's premier national centres for research in astronomy, astrophysics and cosmology.
- Engaged with the IUCAA research community; utilised the institute's library and computational facilities to advance doctoral research on gravitation, modified gravity and relativistic spacetime models.
- Attended seminars and academic discussions with faculty members, post-doctoral fellows and visiting scientists, gaining wider exposure to current developments in cosmology.

Research Scholar | 2023 – Present

Department of P.G. Studies and Research in Mathematics, Kuvempu University, Karnataka, India

- Conducting independent doctoral research on relativistic dynamics of Einstein's field equations and spacetime geometry under expert supervision.
- Investigating wormhole solutions, black hole thermodynamics, and modified gravity models including $f(R)$, $f(R,T)$ and $f(Q,T)$ theories.
- Performing extensive symbolic and numerical computation using Mathematica and Python; visualising and analysing high-dimensional datasets.
- Authoring and co-authoring peer-reviewed research articles in internationally indexed journals (*Annals of Physics*, *Nuclear Physics B*, *Physics Letters B*, *Fortschritte der Physik*, etc.).
- Presenting findings at national and international conferences in mathematics, gravitation and cosmology.
- Engaging in collaborative research with faculty members and fellow scholars; contributing to literature reviews and joint publications.

AWARDS, HONORS & FELLOWSHIPS

- **Qualified Karnataka State Eligibility Test (KSET), 2025** - eligible for Assistant Professorship.
- **Rank-Holder Fellowship** awarded through the National Scholarship Portal (NSP), Government of India, for the academic years **2021–2022**, in recognition of securing First Rank in B.Sc.
- **Selected for the DST INSPIRE Internship Science Camp** — a flagship initiative by the Department of Science and Technology (DST), Government of India, held during **25–29 December 2015** for top-performing science students of Class XI.
- **Gold Medalist** — Highest aggregate marks in B.Sc. (Sahyadri Science College, 2020).
- **Gold Medalist** — Highest marks in Mathematics, M.Sc. (2020).
- **Second Rank** in M.Sc. Mathematics, Kuvempu University (2022).
- **Cash Award** — Highest marks in Mathematics (Fluid Mechanics), B.Sc. (2022).
- **First Prize** — State-level seminar presentation on “Fibonacci Series.”
- **First Prize** — National Science Day seminar presentation.

PUBLICATIONS

Total Publications: 13 (**10 Published + 3 Under Review / Communicated**)

Indexing: SCI / SCIE / Scopus | **Citations / h-index:** 23 / 3

Featured Journals: *Annals of Physics*, *Nuclear Physics B*, *Physics Letters B*, *Fortschritte der Physik*, *Indian Journal of Physics*, *International Journal of Geometric Methods in Modern Physics* and *Progress of Theoretical and Experimental Physics*.

Published / Accepted in Peer-Reviewed Journals

1. **C. S. Varsha**, V. Venkatesha, N. S. Kavya, D. V. Singh, "Impact of EUP correction on thermodynamics of AdS black hole," *Annals of Physics*, 170002, 2025. (Elsevier, Q1, IF – 3.0)
2. N. S. Kavya, **C. S. Varsha**, L. Sudharani, V. Venkatesha, "Unifying non-commutative geometry with Casimir energy: A novel $f(R)$ wormhole solution," *Nuclear Physics B*, 2025. (Elsevier, Q1, IF – 2.8)
3. **C. S. Varsha**, L. Sudharani, N. S. Kavya, V. Venkatesha, "Testing Curvature–Matter Coupling Gravity via Swampland Conjectures," *Fortschritte der Physik*, 2400160, 2025. (Wiley, Q1, IF – 7.8)
4. **C. S. Varsha**, L. Sudharani, N. S. Kavya, V. Venkatesha, "Thermodynamic insights into evolving Lorentzian wormholes in $f(R, T)$ gravity," *Physics Letters B*, 139590, 2025. (Elsevier, Q1, IF – 4.5)
5. **C. S. Varsha**, L. Sudharani, N. S. Kavya, V. Venkatesha, "A novel theoretical approach to dark matter supported Casimir wormhole," *Nuclear Physics B*, 116929, 2025. (Elsevier, Q1, IF – 2.8)
6. V. Venkatesha, G. N. Lathakumari, **C. S. Varsha**, "Noncommutative wormholes and conformal symmetry: a study in $f(Q, T)$ gravity," *Indian Journal of Physics*, 99(13), 5301–5313, 2025. (Springer, Q3, IF – 1.7)
7. **C. S. Varsha**, V. Venkatesha, G. N. Lathakumari, "Cosmic Evolution Under $f(Q, T)$ Gravity: An Energy Conditions Analysis," *International Journal of Geometric Methods in Modern Physics*, 2025. (World Scientific, Q1, IF – 2.2)
8. L. Sudharani, **C. S. Varsha**, V. Venkatesha, "Phase space analysis of Barrow holographic dark energy," *Nuclear Physics B*, 117442, 2026. (Elsevier, Q1, IF – 2.8)
9. K. Suhasini, G. R. Kiran, N. S. Kavya, **C. S. Varsha**, V. Venkatesha, "Geometric aspects of noncommutative wormholes with conformal symmetry," *Physica Scripta*, 101, 18, 2026. (IOPscience, Q2, IF – 2.6)
10. **C. S. Varsha**, L. Sudharani, S. R. Harshitha, & V. Venkatesha, "Gravastar Configurations in Rastall Gravity with Quintessence Effect under the Krori–Barua Metric." *Accepted in Progress of Theoretical and Experimental Physics*, 2026. (Oxford University Press, Q1, IF – 8.3)

Manuscripts Under Review / Communicated

1. S. R. Harshitha, **C. S. Varsha**, & V. Venkatesha, "Casimir Supported Charged Gravastars in Modified Teleparallel Gravity." (*Communicated*), 2026.
2. **C. S. Varsha**, L. Sudharani, S. R. Harshitha, & V. Venkatesha, "Probing Exotic Matter in Gravastar Shells within $f(Q, L_m)$ Gravity: Chaplygin Gas versus MIT Bag Quark Matter." (*Communicated*), 2026.
3. K. Suhasini, G. R. Kiran, N. S. Kavya, **C. S. Varsha**, & V. Venkatesha, "Traversable Wormhole Solutions in $f(Q, L_m)$ Gravity." *arXiv preprint arXiv:2601.08888*, (*Communicated*), 2026.

CONFERENCE PRESENTATIONS

1. "EUP affected thermodynamics and stability of Einstein–Gauss–Bonnet black hole in 4D AdS spacetime," *International Conference on Gravitation, Astrophysics and Cosmology (ICGAC-2024)*, in collaboration with The Tensor Society, India, Mathura, Uttar Pradesh, 14–16 June 2024.
2. "EUP affected thermodynamics and stability of Einstein–Gauss–Bonnet black hole in 4D AdS spacetime," *International Conference on Mathematical Analysis and its Applications (ICMAA)*, Dharwad, Karnataka, 2–4 August 2024.
3. "Thermodynamic Insight into the Evolving Lorentzian Wormhole in $f(R, T)$ Gravity," *International Conference on Astrophysics, Gravitation and Cosmology (APGC-2025)*, in collaboration with GLA University, Mathura, 26–28 June 2025.
4. "Testing Curvature–Matter Coupling Gravity via Swampland Conjectures," *91st Annual Conference of the Indian Mathematical Society – An International Meet*, Department of Mathematics and Astronomy, University of Lucknow, 26–29 December 2025.
5. Invited talk on "Research in Astronomy: Opportunities and Challenges," *Southern Regional Astronomy Meeting (RAM-XI)*, CHRIST (Deemed to be University), Bengaluru, 5–7 February 2026.

CONFERENCES, SEMINARS & WORKSHOPS ATTENDED

- Attended multiple **National and International Conferences and Seminars** in the broad areas of Mathematics, Gravitation, Astrophysics and Cosmology.
- Workshop on **A Python – Powered Workshop on Linear Algebra & Numerical Methods** - Certified.
- Workshop on **Microsoft Office (MS Office) Productivity Tools** - Certified.
- **DST INSPIRE Internship Science Camp** - An initiative by the Department of Science and Technology, Government of India, **25–29 December 2016**; awarded a certificate of participation.

EXTRACURRICULAR & LEADERSHIP EXPERIENCE

Volunteer, Pathways Training & Placement Cell, Sahyadri Science College, Shivamogga (2017 – 2020) — assisted in organizing workshops and student development programmes, delivered presentations to student audiences, and supported placement drives; developed leadership, public-speaking, event-management, and interpersonal skills.

TECHNICAL & PEDAGOGICAL SKILLS

- **Programming Languages:** Python, MATLAB, C++.
- **Scientific Software & Tools:** Wolfram Mathematica, LaTeX (technical writing & typesetting), MS Office.
- **Analytical:** Symbolic computation, data analysis, mathematical modelling, data visualisation.
- **Research:** Literature review, problem formulation, manuscript writing, peer-review responses.
- **Teaching & Communication:** Classroom presentation, lecture preparation, board work, student mentoring, scientific writing.
- **Soft Skills:** Leadership, public speaking, event management, teamwork, organizational and coordination skills.
- **Languages Known:** English (Professional), Kannada (Mother Tongue), Hindi (Basic).

ACADEMIC PROJECTS

M.Sc. Mathematics | Fourth Semester

"A Study on the General Theory of Relativity and its Applications to Wormhole"

- Derived a solution to Einstein's field equations and analysed the energy conditions of wormhole geometries.

B.Sc. Dissertation | Sixth Semester

"Stealth Technology: The Art of Black Magic"

- Studied the principles of stealth technology and its applications in modern defence systems.

PERSONAL DETAILS

- **Father's Name:** Somashekhar
- **Mother's Name:** Prabha
- **Date of Birth:** 17 January 2000
- **Nationality:** Indian
- **Marital Status:** Unmarried
- **Address:** Joseph Nagara, 1st Cross, behind Ather Scooters, near Bank of Baroda, B. H. Road, Shimoga – 577201, Karnataka, India.

DECLARATION

I hereby declare that the above information is true and correct to the best of my knowledge and belief.

Place: Shimoga, Karnataka

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